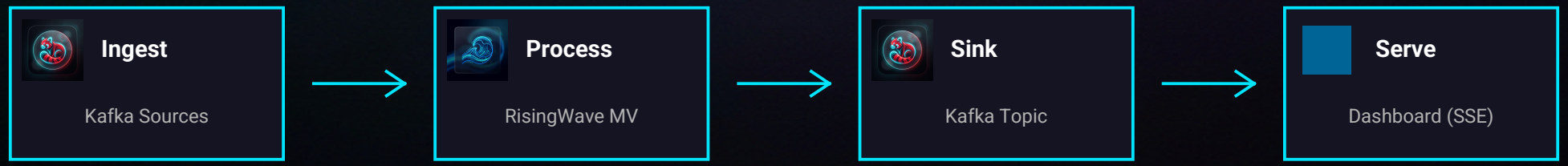
Real-Time Push Architecture

Modern Dashboard via Kafka Sink

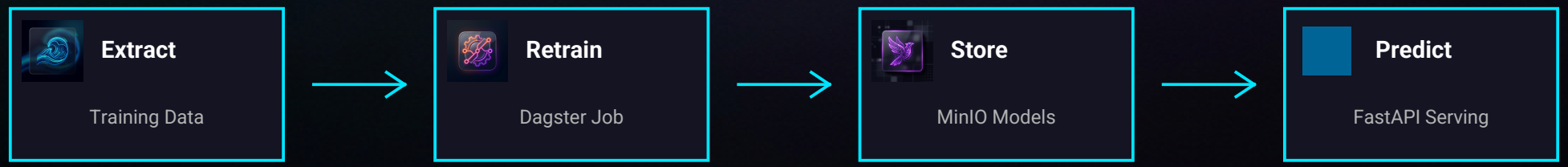
- Decoupled Design: Backend never queries RisingWave directly.
- Real-Time Stream: Data flows via Kafka Sink to Dashboard via SSE.
- Scalability: Buffering ensures dashboard resilience during outages.



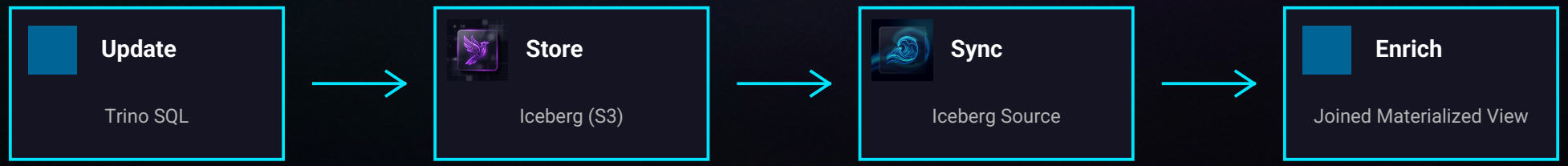
Predictive Analytics Pipeline

Automated ML Training & Hot-Reload Serving

- Continuous Training: retrains every 20 seconds via Dagster.
- Versioned Registry: MinIO stores ML models securely.
- Hot-Reload: FastAPI service reloads models without downtime.



Unified Batch & Stream

Historical Iceberg Integration in RisingWave

- Instant Sync: Iceberg updates are immediately visible in RisingWave.
- Enriched Analytics: Joining event streams with deep historical data.
- Native Connector: High-performance access via REST catalog.